

● MEDIUM

● INFORMATION AND IDEAS

Information and Ideas

16

10 questions · ~15 min

Survey Results for Two Online Account Sign-in Methods

Sign-in method	Percent of participants in the UK who chose method	Percent of participants in Japan who chose method	Percent of participants in India who chose method
Biometrics (for example, a face scan)	33	29	22
Onetime passcodes	16	8	25

A survey listed methods for signing into online accounts. Participants in different countries were asked to choose the sign-in method they view as most secure. The table presents data for two of the methods. According to the table, onetime passcodes were viewed as most secure by _____ blank

Which choice most effectively uses data from the table to complete the text?

- A. 33 percent of survey participants in the UK.
- B. 22 percent of survey participants in India.
- C. 8 percent of survey participants in Japan.
- D. 16 percent of survey participants in India.

Many music streaming services also function as social media platforms: by giving users the option to follow friends and share curated playlists or information about their listening history, these platforms allow people to convey their music preferences and listening activities directly to their social networks. In a 2023 study, researcher Michael James Walsh interviewed frequent users of a popular music streaming platform to investigate how its social media dimensions shape their listening habits. Walsh found that these dimensions tended to make study participants feel more mindful of how their listening activities may be perceived, which in turn influenced how they managed those activities.

Which quotation from a music streaming platform user would best illustrate Walsh's finding?

- A. "When I create playlists or choose songs to listen to, I often pick songs that I think the friends who are following me might be interested in listening to, rather than what I might actually prefer listening to in that moment."
- B. "Because the predictions the platform makes about what new songs I might like depend on my past listening habits, at times its recommendations can feel overly repetitive and unvarying."
- C. "The social aspects of music streaming make it really easy to create a sense of a shared listening experience with my friends, even if we aren't in the same room."
- D. "Listening to music through a streaming platform feels a lot more passive, since I don't need to make any deliberate choices about what music I'm going to consume, like I do when buying digital downloads or records and CDs."

Male túngara frogs make complex calls to attract mates, but their calls also attract frog-biting midges, insects that feed on the frogs' blood. Researchers Ximena Bernal and Priyanka de Silva wondered if the calls alone are sufficient for midges to locate the frogs or if midges use carbon dioxide emitted by frogs as an additional cue to their prey's whereabouts, like mosquitoes do. In an experiment, the researchers placed two midge traps in a túngara frog breeding area. One trap played recordings of túngara frog calls and the other released carbon dioxide along with playing the calls. Bernal and de Silva concluded that carbon dioxide does not serve as an additional cue to frog-biting midges.

Which finding from the experiment, if true, would most directly support Bernal and de Silva's conclusion?

- A. Only a small number of midges were found in the traps, though the majority were found in the trap that played calls and released carbon dioxide.
- B. Midges entered the trap that released carbon dioxide and played calls only during or immediately after periods of carbon dioxide release.
- C. More midges were found in the trap that only played calls than in the trap that played calls and released carbon dioxide.
- D. The trap that released carbon dioxide and played calls attracted few midges when carbon dioxide concentrations were low but attracted many midges when carbon dioxide concentrations were high.

E-book Sales as a Percentage of Total Unit Sales in All Book Formats for a Large US Trade Publisher, by Genre, 2006, 2011, 2016

Genre	2006	2011	2016
science fiction and fantasy	0.6	27.7	36.7
cookbooks	0	2.9	10.5
travel guides	0	5.5	24.6
romance	0.3	40.6	56.2

E-books became an increasingly popular means of reading in the United States in the 2000s and 2010s, though that popularity was concentrated in titles that, like those in most fiction genres, are meant to be read straight through from beginning to end. For books in nonfiction genres that do not tell stories and require the reader to flip back and forth through a volume, e-books were significantly less commercially successful. This can be seen by comparing _____blank

Which choice most effectively uses data from the table to illustrate the claim?

- A. the percentage of 2016 cookbook sales that were e-books with the percentage of 2016 science fiction and fantasy sales that were e-books.
- B. the percentage of 2006 romance sales that were e-books with the percentage of 2016 romance sales that were e-books.
- C. the percentage of 2006 romance sales that were e-books with the 2006 science fiction and fantasy sales that were e-books.

- D. the percentage of 2011 travel guide sales that were e-books with the percentage of 2016 travel guide sales that were e-books.

Although many transposons, DNA sequences that move within an organism's genome through shuffling or duplication, have become corrupted and inactive over time, those from the long interspersed nuclear elements (LINE) family appear to remain active in the genomes of some species. In humans, they are functionally important within the hippocampus, a brain structure that supports complex cognitive processes. When the results of molecular analysis of two species of octopus—an animal known for its intelligence—were announced in 2022, the confirmation of a LINE transposon in *Octopus vulgaris* and *Octopus bimaculoides* genomes prompted researchers to hypothesize that that transposon family is tied to a species' capacity for advanced cognition.

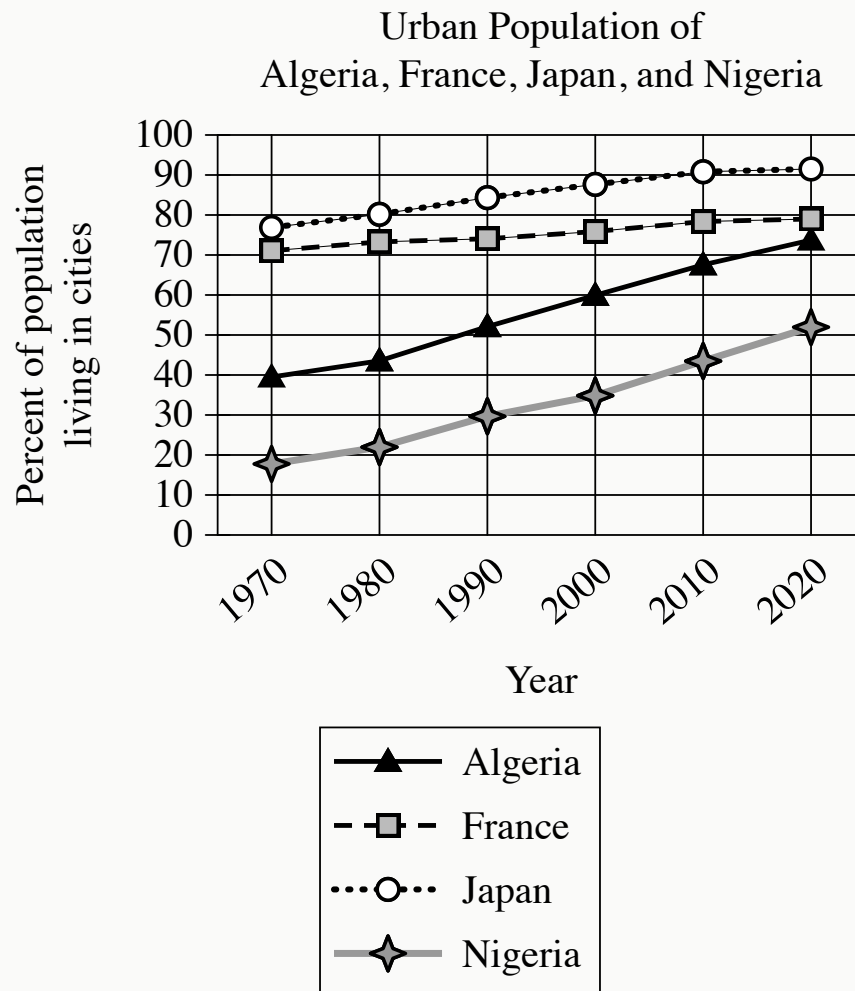
Which finding, if true, would most directly support the researchers' hypothesis?

- A. The LINE transposon in *O. vulgaris* and *O. bimaculoides* genomes is active in an octopus brain structure that functions similarly to the human hippocampus.
- B. The human genome contains multiple transposons from the LINE family that are all primarily active in the hippocampus.
- C. A consistent number of copies of LINE transposons is present across the genomes of most octopus species, with few known corruptions.
- D. *O. vulgaris* and *O. bimaculoides* have smaller brains than humans do relative to body size, but their genomes contain sequences from a wider variety of transposon families.

Fish whose DNA has been modified to include genetic material from other species are known as transgenic. Some transgenic fish have genes from jellyfish that result in fluorescence (that is, they glow in the dark). Although these fish were initially engineered for research purposes in the 1990s, they were sold as pets in the 2000s and can now be found in the wild in creeks in Brazil. A student in a biology seminar who is writing a paper on these fish asserts that their escape from Brazilian fish farms into the wild may have significant negative long-term ecological effects.

Which quotation from a researcher would best support the student's assertion?

- A. "In one site in the wild where transgenic fish were observed, females outnumbered males, while in another the numbers of females and males were equivalent."
- B. "Though some presence of transgenic fish in the wild has been recorded, there are insufficient studies of the impact of those fish on the ecosystems into which they are introduced."
- C. "The ecosystems into which transgenic fish are known to have been introduced may represent a subset of the ecosystems into which the fish have actually been introduced."
- D. "Through interbreeding, transgenic fish might introduce the trait of fluorescence into wild fish populations, making those populations more vulnerable to predators."



- The following 4 lines are shown:
 - Algeria
 - France
 - Japan
 - Nigeria
- The Algeria line:
 - Begins at 1970, 39.5 percent
 - Rises gradually to 1980, 43.54 percent
 - Rises sharply to 1990, 52.09 percent
 - Rises gradually to 2000, 59.92 percent

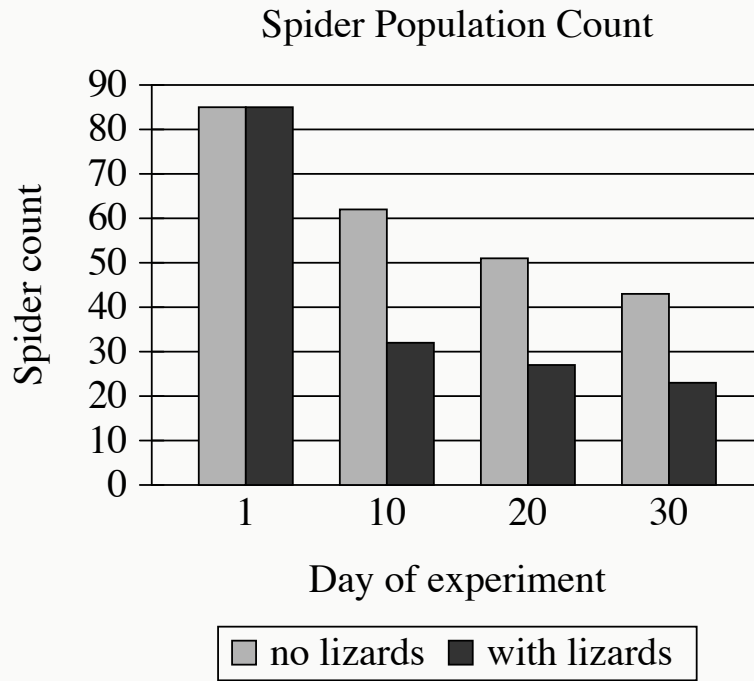
- Rises gradually to 2010, 67.54 percent
- Rises gradually to 2020, 73.73 percent
- The France line:
 - Begins at 1970, 71.06 percent
 - Rises gradually to 1980, 73.28 percent
 - Rises gradually to 1990, 74.06 percent
 - Rises gradually to 2000, 75.87 percent
 - Rises gradually to 2010, 78.37 percent
 - Rises gradually to 2020, 79 percent
- The Japan line:
 - Begins at 1970, 76.88 percent
 - Rises gradually to 1980, 80.17 percent
 - Rises gradually to 1990, 84.34 percent
 - Rises gradually to 2000, 87.65 percent
 - Rises gradually to 2010, 90.81 percent
 - Rises gradually to 2020, 91.5 percent
- The Nigeria line:
 - Begins at 1970, 17.76 percent
 - Rises gradually to 1980, 21.97 percent
 - Rises gradually to 1990, 29.68 percent
 - Rises gradually to 2000, 34.84 percent
 - Rises sharply to 2010, 43.48 percent
 - Rises sharply to 2020, 51.96 percent

The share of the world's population living in cities has increased dramatically since 1970, but this change has not been uniform. France and Japan, for example, were already heavily urbanized in 1970, with 70% or more of the population living in cities. The main contributors to the world's urbanization since 1970 have been countries like Algeria, whose population went from _____blank

Which choice most effectively uses data from the graph to complete the assertion?

- A. around 50% urban in 1970 to around 90% urban in 2020.
- B. less than 40% urban in 1970 to around 90% urban in 2020.
- C. less than 20% urban in 1970 to more than 50% urban in 2020.

D. around 40% urban in 1970 to more than 70% urban in 2020.



- For each data category, the following bars are shown:
 - no lizards
 - with lizards
- All values are approximate.
- The data for the 4 categories are as follows:
 - Day 1:
 - no lizards: 85
 - with lizards: 85
 - Day 10:
 - no lizards: 62
 - with lizards: 32
 - Day 20:
 - no lizards: 51
 - with lizards: 27
 - Day 30:
 - no lizards: 43
 - with lizards: 23

To investigate the effect of lizard predation on spider populations, a student in a biology class placed spiders in two enclosures, one with lizards and one without, and tracked the number of spiders in the enclosures for 30 days. The student concluded that the reduction in the spider population count in the enclosure with lizards by day 30 was entirely attributable to the presence of the lizards.

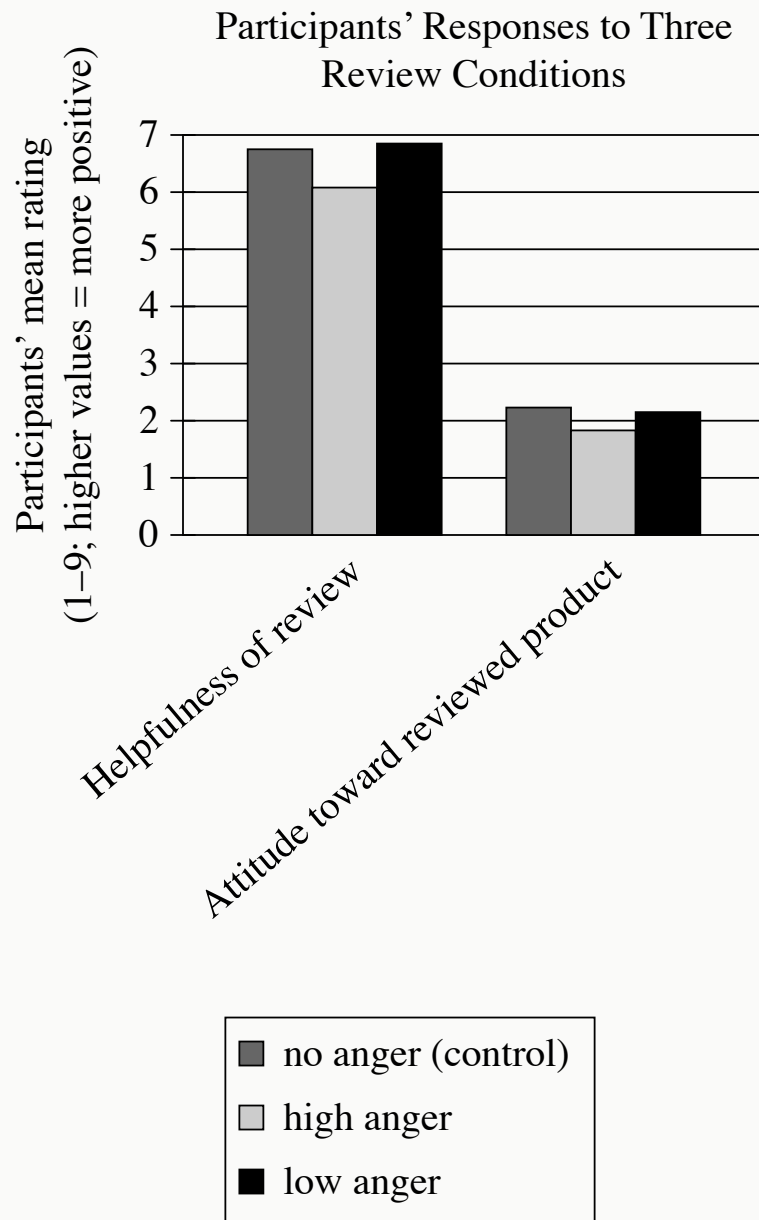
Which choice best describes data from the graph that weaken the student's conclusion?

- A. The spider population count was the same in both enclosures on day 1.
- B. The spider population count also substantially declined by day 30 in the enclosure without lizards.
- C. The largest decline in spider population count in the enclosure with lizards occurred from day 1 to day 10.
- D. The spider population count on day 30 was lower in the enclosure with lizards than in the enclosure without lizards.

Plants like potatoes, tomatoes, and soybeans are susceptible to bacterial wilt disease caused by the bacteria *Ralstonia solanacearum*. A multinational team of scientists led by Zhong Wei studied whether other microbes in the soil might influence the degree to which plants are affected by the disease. The team sampled soil surrounding individual tomato plants over time and compared the results of plants that became diseased with those that remained healthy. They concluded that the presence of certain microbes in the soil might explain the difference between healthy and diseased plants.

Which finding, if true, would most directly support the team's conclusion?

- A. The soil surrounding healthy plants contained significantly higher concentrations of microbes known to inhibit *Ralstonia solanacearum* than the soil surrounding diseased plants did.
- B. The soil surrounding the plants contained high concentrations of *Ralstonia solanacearum* regardless of whether the plants were affected by wilt disease.
- C. The soil surrounding healthy plants tended to have significantly higher moisture levels than the soil surrounding diseased plants did.
- D. By the end of the experiment, over half the plants had been affected by wilt disease regardless of differences in the types and concentrations of microbes in the surrounding soil.



- For each data category, the following bars are shown:
 - no anger (control)
 - high anger
 - low anger
- The data for the 2 categories are as follows:
 - Helpfulness of review:

- no anger (control): 6.75
- high anger: 6.08
- low anger: 6.85
- Attitude toward reviewed product:
 - no anger (control): 2.23
 - high anger: 1.83
 - low anger: 2.15

To understand how expressions of anger in reviews of products affect readers of those reviews, business scholar Dezhi Yin and colleagues measured study participants' responses to three versions of the same negative review—a control review expressing no anger, a review expressing a high degree of anger, and a review expressing a low degree of anger. Reviewing the data, a student concludes that the mere presence of anger in a review may not negatively affect readers' perceptions of the review, but a high degree of anger in a review does worsen readers' perceptions of the review.

Which choice best describes data from the graph that support the students' conclusion?

- A. On average, participants' ratings of the helpfulness of the review were substantially higher than were participants' ratings of the reviewed product regardless of which type of review participants had seen.
- B. Compared with participants who saw the control review, participants who saw the low-anger review rated the review as slightly more helpful, whereas participants who saw the high-anger review rated the review as less helpful.
- C. Participants who saw the low-anger review rated the review as slightly more helpful than participants who saw the control review did, but participants' attitude toward the reviewed product was slightly worse when participants saw the low-anger review than when they saw the no-anger review.
- D. Compared with participants who saw the low-anger review, participants who saw the high-anger review rated the review as less helpful and had a less positive attitude toward the reviewed product.